



The GNU Build System

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The GNU Build System

- A complete package for automating software builds
- Includes auto tools for checking system info, setting up makefiles, handling preprocessing, compilation, linking and installation of files, and managing library and object file dependencies

The GNU Build System

- Consists of several parts:
 - Collectively, Autotools:
 - Automake
 - Autoconf
 - Libtool
 - GNU make
 - Can also add the linux command “install”, though not usually listed as part of the GNU build system

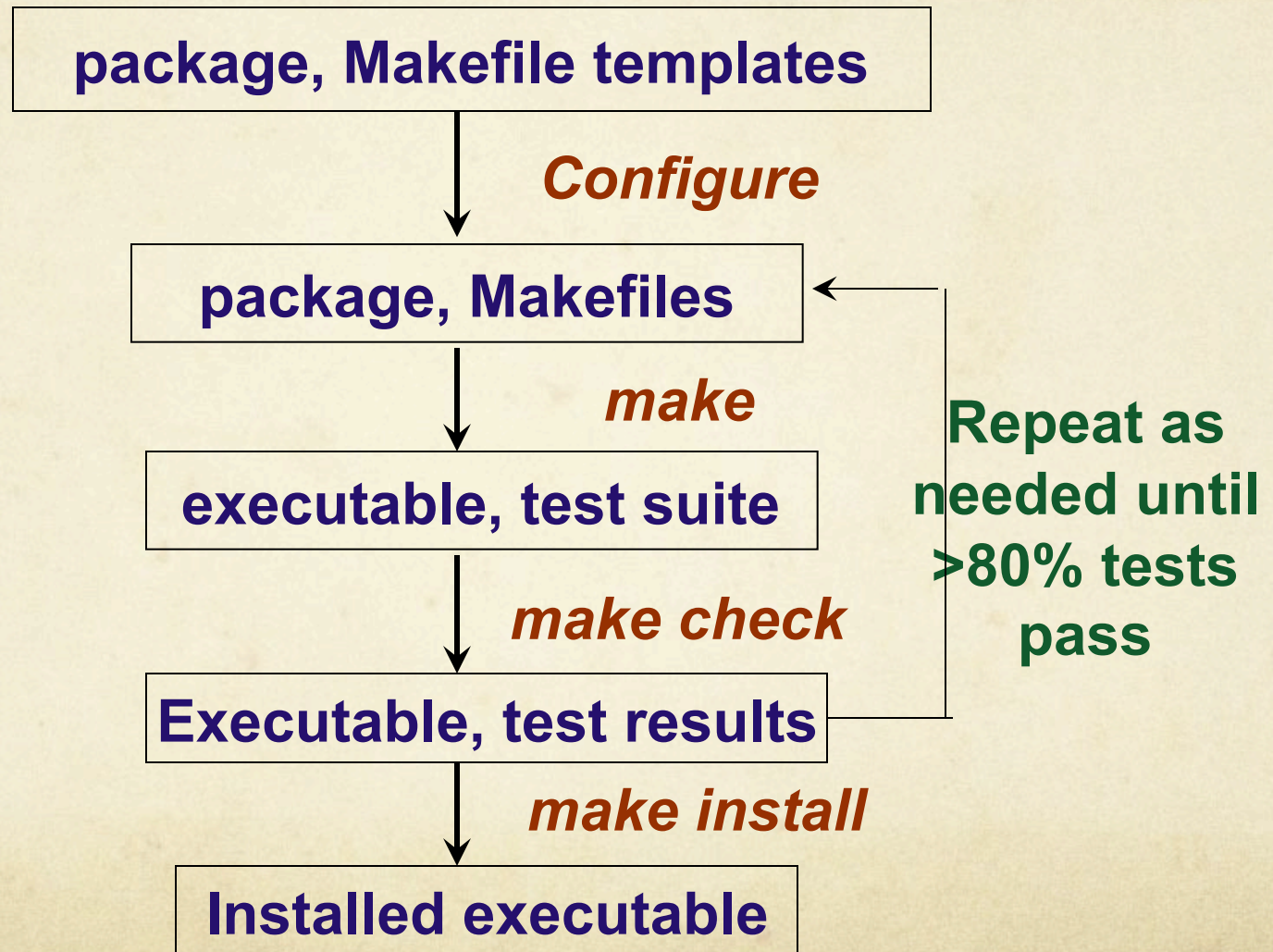
The GNU Build System

- Ties together:
 - System setup (location of libraries, include files, testing compilers and options, etc.)
 - Building Makefiles from templates
 - Preprocessing (with compilers or CPP)
 - Compiling
 - Linking
 - Testing (if test suite is made available)
 - installation

The GNU Build System

- Why use the GNU build system?
 - Portability can be built in with preprocessor directives and the help of autoconf
 - Larger package builds can be easily automated, especially after changes
 - *much* easier for the end user to build a package since there are only four little commands to run...

The “4-Step” Build on Linux



A Very Brief Intro to Configure Scripts

- Configure automates system setup and generation of makefiles
- Automatic makefile generation done with automake
- Can be built with autoconf or “home-grown”
- Normally use autoconf

Configure Scripts

- Sets up things like:
 - Compilers
 - Include files
 - Libraries (usually external)
 - Preprocessor info
 - Where to install built binaries
 - Any compiler/linker/preprocessor flags

Configure Scripts

- These are added to a makefile template and the makefile is built
- Use “M4” to build macros for setting up makefiles
- Built on top of the Bourne shell (sh)
- ... however they are complex enough that knowledge of sh is not enough to get you started making your own

Example Configure Run

- For this course, you don't need to know how to build configure scripts
- ... but it is good for you to see them in action and go through the pain of using one
- Keep in mind that they do have a serious downside:
 - They are only as good as their builder made them to be...

The rest of it...

- So, don't worry about how configure goes from a system setup to a set of usable makefiles
- For those interested, the automake tool automates generation of makefiles
- ... and libtool handles library and object dependencies

The rest of it...

- Most of what we will talk about is:
 - Preprocessors
 - Compilers
 - Linkers, executable generation
 - GNU make
- This will do for any budding programmer...